significantly. Hence in modern diesel engines this system is used for good fuel economy and lower emission levels. In this common rail direct injection system the fuel pressure is maintained at 1350 – 2000 bar in the common rail by using a fuel pump. The fuel is supplied to all the injectors from the common rail and injected into the combustion chamber according to the firing order. The pressure maintained in the common rail is not depending on the engine's operation. Hence the CRDI system shows as better method as compared to the conventional engine systems. In the CRDI system the common rail and the high pressure lines connected with the injectors are designed to withstand very high fuel pressures. The electronic control unit (called as ECU) present in the system determines the amount of diesel fuel injected, the fuel pressure, injection timing and the injection duration. The ECU receives the input signals from different sensors, calculates the amount of fuel required, time for injection duration and other details for injection of the fuel and delivers required output signals to the solenoid operated injectors. By this way the fuel injection pressure, time and duration of injection and the amount of fuel injected are controlled.

Construction of CRDI system: The CRDI system consists of a fuel lift pump, a high pressure diesel pump, a common rail for storing diesel at high pressure and solenoid operated diesel injectors. In addition the system has different sensors, actuators and an electronic control unit (ECU). The lift pump is used to pump the diesel fuel through the filter to the high pressure diesel pump. The diesel from the high pressure pump transfers the fuel to the common rail. The injectors mounted on the cylinder head of the engine receive the high pressure fuel from the common rail through the high pressure lines. The solenoid operated injectors open the injector holes according to the signals received from the ECU. 16 bit or 32 bit microprocessor acts as the brain of the ECU used in the CRDI system. The temperature of the engine, temperature of the air and fuel, the rail pressure, engine booster pressure, accelerator pedal position, vehicle speed and intake air amount etc. are monitored continuously by the ECU using different sensors located at different parts of the engine by input signals. Figure 9.12(a) presents the construction of the CRDI system

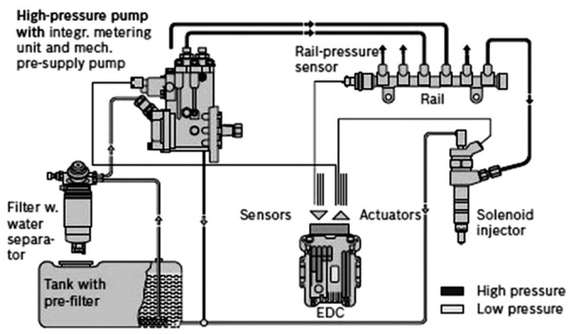


Fig 9.12(a) Construction of the CRDI system

Operation: Diesel fuel from the fuel tank is lifted by the fuel lift pump and supplied to the fuel filter. The fuel lift pump is operated electrically without the support of the engine power. The high pressure pump operated by the engine develops very high pressure and forces the high pressure fuel to the common rail. The fuel from the common rail is supplied to the individual injectors through the high pressure lines and the fuel is sprayed in the cylinder at very high pressure. The electronic control unit receives the

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